

1. Administrative & Study Identification

Official Study Title: A Phase 3, Open-label, Randomized Study of Nivolumab Combined With Ipilimumab, or With Standard of Care Chemotherapy, Versus Standard of Care Chemotherapy in Participants With Previously Untreated Unresectable or Metastatic Urothelial Cancer, **Public Study Title:** Study of Nivolumab in Combination With Ipilimumab or Standard of Care Chemotherapy Compared to the Standard of Care Chemotherapy Alone in Treatment of Participants With Untreated Inoperable or Metastatic Urothelial Cancer **Short Title/Acronym:** CheckMate 901, **Protocol Number:** CA209-901, **NCT Number:** NCT03036098, **Posting ID:** 177882007837354208 **Sponsor Name:** Bristol-Myers Squibb, **Investigational Product:** Nivolumab (Opdivo) and Ipilimumab (Yervoy), **Phase of Development:** Phase 3, **Medical Monitor:** Bristol-Myers Squibb Clinical Operations / Medical Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine whether the investigational immunotherapy nivolumab in combination with ipilimumab, or nivolumab in combination with standard of care (SOC) chemotherapy, is more effective than SOC chemotherapy alone in improving Overall Survival (OS) and Progression-Free Survival (PFS) in treating participants with previously untreated inoperable or metastatic urothelial cancer. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DOR), and safety/tolerability profiles associated with the treatment arms.

2.2 Background & Rationale To address the critical unmet need in previously untreated unresectable or metastatic urothelial carcinoma (mUC). Platinum-based chemotherapy regimens (like cisplatin and gemcitabine) have been the standard of care for mUC for decades, but durable responses are rare. Furthermore, many patients are ineligible for cisplatin. Nivolumab has previously demonstrated durable responses in patients with mUC progressing despite platinum-based chemotherapy. The CheckMate 901 trial was designed to evaluate if incorporating nivolumab in the first-line setting—either with CTLA-4 inhibitor ipilimumab or with SOC chemotherapy—could provide significant OS and PFS benefits over standard chemotherapy alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional, **Phase:** PHASE3
Trial Status: ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Control Type:** Active Comparator (Standard of Care Chemotherapy: Gemcitabine + Cisplatin/Carboplatin), **Blinding:** Open-label, **Randomization:** Randomized (1:1), stratified by tumor PD-L1 expression and presence of liver metastasis.

3.2 Planned Enrollment & Duration Target \$N\$: 1,314 participants (Actual Enrollment) **Number of Sites:** Global, Multicenter, **Estimated Study Start:** 24-Mar-2017 **Estimated Primary Completion:** 30-Aug-2024 **Estimated Study Completion:** 15-May-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histological or cytological evidence of metastatic or surgically inoperable transitional cell cancer (TCC) of the urothelium involving the renal pelvis, ureter, bladder or urethra. 2. No prior systemic chemotherapy for metastatic or surgically inoperable urothelial cancer (UC). 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. Measurable disease by CT or MRI per RECIST 1.1 criteria. 5. Women and men must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Disease that is suitable for local therapy administered with curative intent. 2. Active brain metastases or leptomeningeal metastases. 3. Active, known, or suspected autoimmune disease. 4. Prior treatment with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CD137, or anti-CTLA-4 antibody, or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways. 5. Any serious or uncontrolled medical disorder in the opinion of the investigator that may increase the risk associated with study participation or study drug administration or interfere with the interpretation of study results.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Combination with Chemotherapy Arm:** Nivolumab 360 mg every 3 weeks + Gemcitabine-Cisplatin (Platinol) every 3 weeks for up to 6 cycles, followed by Nivolumab monotherapy 480 mg every 4 weeks (or 240 mg every 2 weeks). * **Dual Immunotherapy Arm:** Nivolumab 1 mg/kg + Ipilimumab 3 mg/kg every 3 weeks for up to 4 doses, followed by Nivolumab 480 mg every 4 weeks. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Combination regimens administered for up to 4-6 cycles; maintenance nivolumab monotherapy continued until disease progression, unacceptable toxicity, or a maximum of 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Carboplatin was permitted for patients discontinuing cisplatin (Platinol). Prior palliative radiotherapy completed at least 2 weeks before study drug administration. **Prohibited:** Live/attenuated vaccines within 30 days prior to study start. Concurrent systemic immunosuppressive therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy evaluation, RECIST 1.1 Imaging
Baseline	Day 0	Randomization, First Dose of Immunotherapy +/- Chemotherapy
Interim	Every 3 to 4 Weeks	Treatment administration, Safety Labs, Efficacy Assessment (Imaging), AE monitoring
End of Study	Up to 2 Years	Final Efficacy Scans, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Overall Survival (OS) in Cisplatin-ineligible Randomized Participants for Primary Study. 2. Overall Survival (OS) in Programmed Death-Ligand 1 (PD-L1) Positive ($\geq 1\%$) Randomized Participants by Immunohistochemistry (IHC) for Primary Study. 3. Progression-Free Survival (PFS) by Blinded Independent Central Review (BICR) [Using Response Evaluation Criteria in Solid Tumors (RECIST) 1.1] in Cisplatin-ineligible Participants for Sub-study. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) using RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) in All Randomized Participants for Primary Study. 2. Progression-Free Survival (PFS) by Blinded Independent Central Review (BICR) [Using RECIST 1.1] in Cisplatin-ineligible Randomized Participants for Primary Study. 3. Progression-Free Survival (PFS) by Blinded Independent Central Review (BICR) [Using RECIST 1.1] in Programmed Death-Ligand 1 (PD-L1) Positive ($\geq 1\%$) Participants for Primary Study. 4. Objective Response Rate (ORR), Duration of Response (DOR), and Complete Response (CR) rates.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring per CTCAE criteria. Key AEs monitored include nausea, fatigue, musculoskeletal pain, and immune-mediated adverse reactions.

Serious Adverse Events (SAEs): Reported promptly. Notable SAEs ($>2\%$) in the trial included urinary tract infection, acute kidney injury, anemia, pulmonary embolism, and sepsis. **Data Monitoring**

Committee (DMC): Independent Data Monitoring Committee (IDMC) periodically reviews safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population utilized for primary efficacy outcomes; Safety Set utilized for all safety/toxicity reporting. **Sample Size Justification:** Powered to detect statistically significant improvements in OS and PFS. At the final analysis, required OS and PFS events were tracked simultaneously based on an event-driven and time-driven approach. **Handling of Missing Data:** Standard oncology censoring rules applied for PFS (e.g., censoring for subsequent anti-cancer therapy prior to progression).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular on-site and remote monitoring visits for source data verification and regulatory compliance checks. **Audit Trail:** Secure eCRF systems deployed with full audit trails and prompt data clarification protocols.

11. Study Identifiers & Governance

Primary ID: {{CA209-901}} **Posting ID:** {{177882007837354208}}
Registry Number: {{NCT03036098}} **Trial Status:** {{ACTIVE_NOT_RECRUITING}} **Sponsor/Lead Organization:** {{Bristol-Myers Squibb}} **Study Title:** {{Study of Nivolumab in Combination With Ipilimumab or Standard of Care Chemotherapy Compared to the Standard of Care Chemotherapy Alone in Treatment of Participants With Untreated Inoperable or Metastatic Urothelial Cancer}}
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12. Clinical Framework (Oncology Specific)

Primary Condition: {{Urothelial Cancer / Metastatic Urothelial Cancer (Preferred Name: Suspected urothelial cancer; Semantic Type: Finding)}} **Condition Definition:** {{Urothelial cancer is a type of malignancy originating from the urothelial cells that line the urinary tract. While most frequently found in the bladder, it can also develop in the renal pelvis, ureters, and urethra, representing the uncontrolled growth of these

